



Grade 5 FAST Mathematics

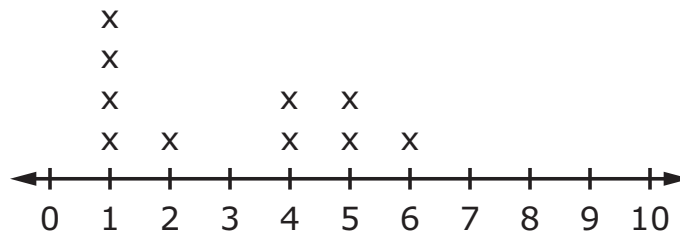
2025 Released Test Items

1. Select all the expressions that represent 158.901.

- Ⓐ 158 ones + 901 tenths
- Ⓑ 158 ones + 901 thousandths
- Ⓒ 158 hundreds + 901 hundredths
- Ⓓ 1 hundred + 58 tens + 9 tenths + 1 thousandth
- Ⓔ 1 hundred + 58 ones + 90 hundredths + 1 thousandth
- Ⓕ 1 hundred + 5 tens + 8 ones + 9 tenths + 1 thousandth

Grade 5 FAST Mathematics

2. A data set is shown on the line plot.



Select the mean and range of the data set.

	1	3	5
Mean	(A)	(B)	(C)
Range	(D)	(E)	(F)

3. What is 437×285 ?

←

→

↶

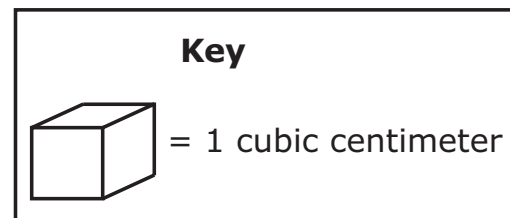
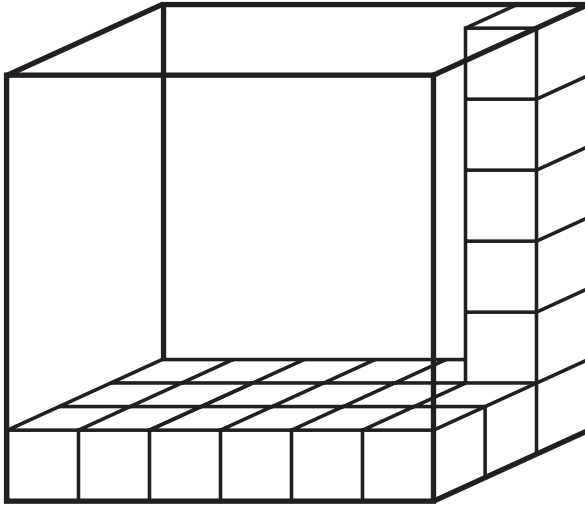
↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

4. This question has **two** parts.

A right rectangular prism is shown. The base is covered with cubes with edge lengths of 1 centimeter.

**Part A**

What is the total volume, in cubic centimeters, of the right rectangular prism?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Part B

How many more cubes would it take to fill the prism?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

5. An input-output table models the expression $3x - 1$.

Input (x)	1	2	4
Output	2	5	11

Select all the columns that could be added to the table.

Ⓐ

Input (x)	7
Output	22

Ⓒ

Input (x)	5
Output	12

Ⓔ

Input (x)	8
Output	23

Ⓑ

Input (x)	3
Output	8

Ⓓ

Input (x)	6
Output	19

- 6.** Select a word and a phrase to complete the sentence describing attributes of a rhombus.

Since a rhombus is always a

- Ⓐ square
Ⓑ rectangle
Ⓒ parallelogram
- ,

it must have

- Ⓐ two long sides and two short sides
Ⓑ four equal angles
Ⓒ two pairs of parallel sides
- .

7. Natalia has 2 packages of pencils. There are 10 pencils in each package. She gives away some pencils, p , to her friends. She has 8 pencils remaining.

Which equation can be used to determine the number of pencils, p , Natalia gives to her friends?

- Ⓐ $(p - 8) \times 2 = 10$
- Ⓑ $10 \times 2 - p = 8$
- Ⓒ $8 \times 2 - 10 = p$
- Ⓓ $10 \times 2 + 8 = p$

8. Katie walks 2 miles. Every $\frac{1}{2}$ mile, she stops to stretch.

Which expression represents how many times Katie stops to stretch on her walk?

Ⓐ $2 \times \frac{1}{2}$

Ⓑ $2 \div \frac{1}{2}$

Ⓒ $2 + \frac{1}{2}$

Ⓓ $2 - \frac{1}{2}$

9. What is $11,570 \div 43$?

Ⓐ $245 \frac{35}{43}$

Ⓑ $260 \frac{39}{43}$

Ⓒ $269 \frac{3}{43}$

Ⓓ $273 \frac{38}{43}$

10. The number 0.307 is incorrectly written in expanded form, as shown.

$$3 \times \frac{1}{10} + 7 \times \frac{1}{100}$$

Which action will correct the error?

- Ⓐ replacing $3 \times \frac{1}{10}$ with $30 \times \frac{1}{10}$
- Ⓑ replacing $3 \times \frac{1}{10}$ with $3 \times \frac{1}{1000}$
- Ⓒ replacing $7 \times \frac{1}{100}$ with $70 \times \frac{1}{10}$
- Ⓓ replacing $7 \times \frac{1}{100}$ with $7 \times \frac{1}{1000}$

Grade 5 FAST Mathematics

11. A partially completed input-output table for the expression $4x$ is shown.

Input (x)	1	3	8	?
Output	4	12	32	48

What is the value of the missing input?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

12. Which comparison is true?

- Ⓐ $367.115 > 376.114$
- Ⓑ $367.115 > 367.105$
- Ⓒ $367.115 > 376.015$
- Ⓓ $367.115 > 367.511$

13. What is the value of $54.2 \div 0.1$?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

14. An expression is shown.

$$\frac{13}{9} - \frac{3}{4}$$

What is the value of the expression?

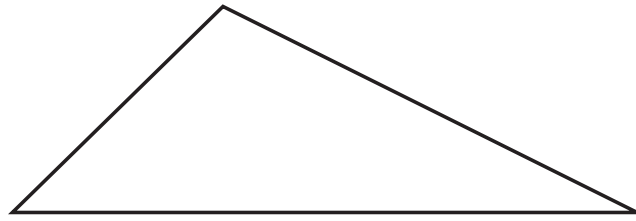
Ⓐ $\frac{10}{5}$

Ⓑ $\frac{25}{5}$

Ⓒ $\frac{10}{36}$

Ⓓ $\frac{25}{36}$

15. A triangle is shown.



Which pair of terms classifies the triangle?

- Ⓐ acute and scalene
- Ⓑ acute and isosceles
- Ⓒ obtuse and scalene
- Ⓓ obtuse and isosceles

- 16.** Susanna selects a recipe that uses $\frac{5}{3}$ teaspoons of salt. She has $\frac{1}{2}$ teaspoon of salt.

How much more salt does Susanna need for the recipe?

- Ⓐ $\frac{4}{1}$ teaspoons
- Ⓑ $\frac{6}{5}$ teaspoons
- Ⓒ $\frac{7}{6}$ teaspoons
- Ⓓ $\frac{13}{6}$ teaspoons

17. What is the value of $81.128 - 51.428$?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

18. A library has 5 large boxes that contain books for a book sale.

- Each box contains 24 hardcover books and 32 paperback books.
- The library sells 77 books on the first day of the book sale.

How many books does the library have left after the first day of the book sale?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

19. Which equation is true?

Ⓐ $4.2 \div (2 + 5) = 1.50 + 1.44$

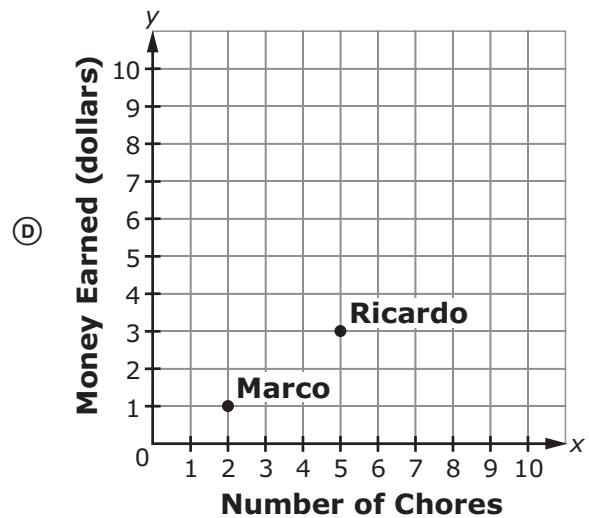
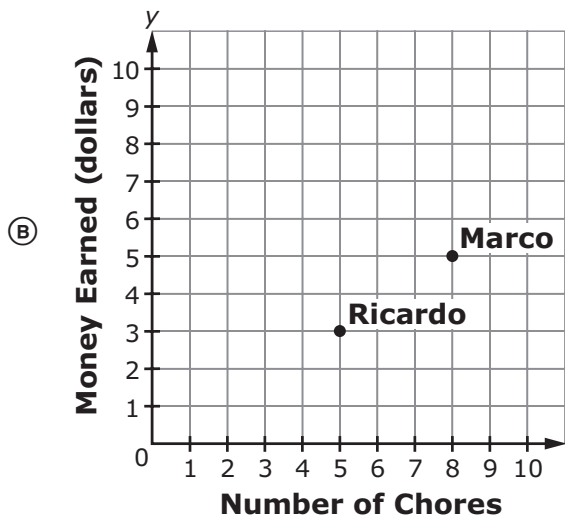
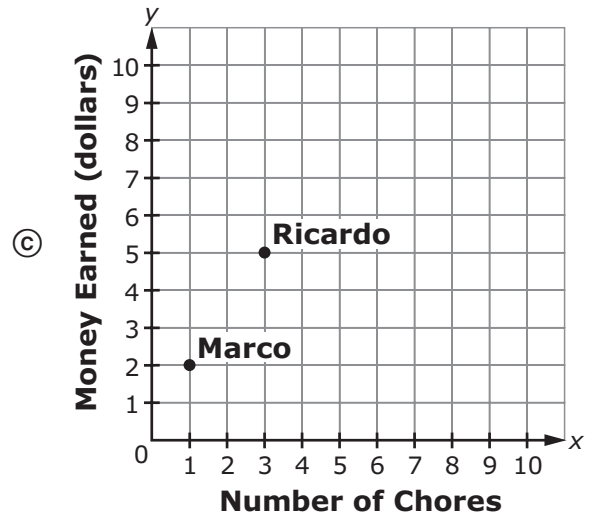
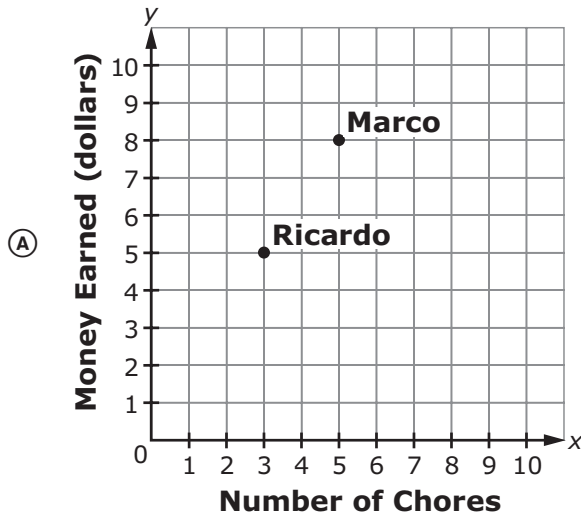
Ⓑ $4.2 \div (2 + 5) = 1.50 - 1.44$

Ⓒ $4.2 \div (2 + 5) = 3.85 + 3.25$

Ⓓ $4.2 \div (2 + 5) = 3.85 - 3.25$

- 20.** Ricardo completes 3 chores and earns 5 dollars. Marco completes 2 more chores and earns 3 more dollars than Ricardo. They plot points on a graph to show how many chores and how much money they each earn.

Which graph shows the points for Ricardo and Marco?



21. An expression is shown.

$$3 + 56 \div 8$$

What is the value of the expression?

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

22. A number is shown in expanded form.

$$(9 \times 100) + (4 \times 10) + (8 \times 1) + \left(2 \times \frac{1}{10}\right) + \left(5 \times \frac{1}{100}\right)$$

What is the number in standard decimal form?

←

→

↶

↷

✖

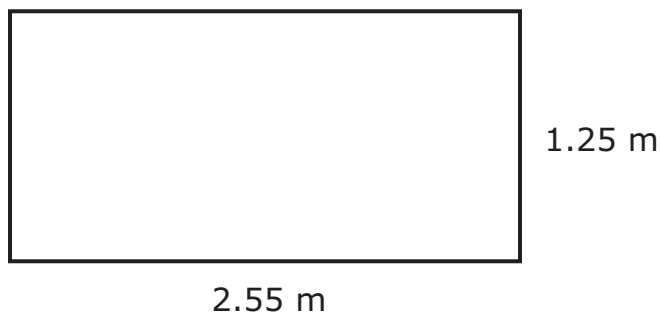
1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

- 23.** A library has 649 fiction books and 506 nonfiction books. All the books are on 11 shelves. Each shelf has the same number of books.

How many books are on each shelf?

- Ⓐ 15
- Ⓑ 46
- Ⓒ 59
- Ⓓ 105

24. The rectangle shown has side lengths in meters (m).



What is the perimeter, in meters, of the rectangle?

← → ↶ ↷ ✕

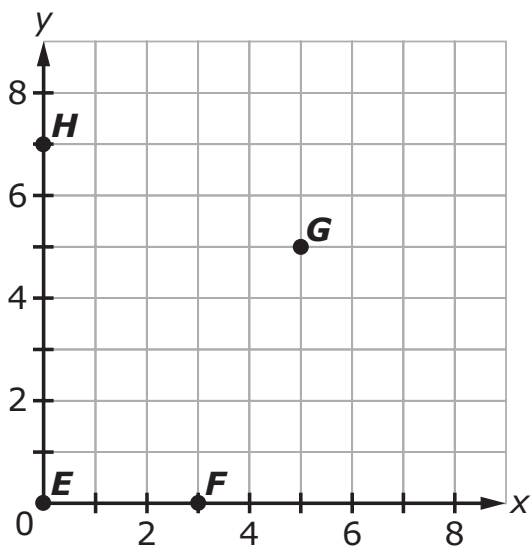
1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

- 25.** Cheryl has $8\frac{2}{5}$ pounds of apples. She uses $4\frac{1}{2}$ pounds to make applesauce and $3\frac{1}{4}$ pounds to make apple pie.
How many pounds of apples does Cheryl have left over?

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

26. Four points are shown on the coordinate plane.



Match each point to its location.

	x-Axis Only	y-Axis Only	Origin	Neither Axis nor Origin
point E	(A)	(B)	(C)	(D)
point F	(E)	(F)	(G)	(H)
point G	(I)	(J)	(K)	(L)
point H	(M)	(N)	(O)	(P)

27. Which equation is true?

- Ⓐ $5 + 3 \times 4 = 4 \times 3 + 5$
- Ⓑ $5 + 3 \times 4 = 5 \times 3 + 4$
- Ⓒ $5 + 3 \times 4 = (5 + 3) \times 4$
- Ⓓ $5 + 3 \times 4 = 5 \times 4 + 3 \times 4$

28. Select all the numbers that have a value of 38.5 when rounded to the nearest tenth.

- Ⓐ 38.518
- Ⓑ 38.464
- Ⓒ 38.553
- Ⓓ 38.417
- Ⓔ 38.507

29. What is $2,372 \times 431$?

- Ⓐ 889,232
- Ⓑ 900,332
- Ⓒ 1,022,332
- Ⓓ 10,223,320

30. What is the value of $\frac{30}{12} + \frac{4}{8}$?

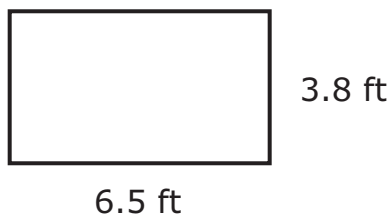
Ⓐ $\frac{34}{24}$

Ⓑ $\frac{34}{20}$

Ⓒ $\frac{72}{24}$

Ⓓ $\frac{38}{12}$

31. A rectangle is shown, with side lengths in feet (ft).



Complete the sentences to show the perimeter and area of the rectangle.

The perimeter is

feet.

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
0	.	$\frac{\square}{\square}$		

The area is

square feet.

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
0	.	$\frac{\square}{\square}$		

32. A statement is shown.

“The sum of 37 and the product of 5 and p is 97.”

Which equation represents the statement?

Ⓐ $37 + (5 \times p) = 97$

Ⓑ $37 \times (5 + p) = 97$

Ⓒ $(37 + 5) \times p = 97$

Ⓓ $(37 \times 5) + p = 97$

33. Alex writes the pattern shown.

5, 8, 11, 14, . . .

He wants to create a rule that describes the pattern.

Select terms to complete the statement describing a rule for Alex's pattern.

A rule that describes the pattern is [☐ A $2x$ ☐ B $3x$ ☐ C $5x$] +
[☐ A 0 ☐ B 2 ☐ C 3 ☐ D 5], where $x = 1, 2, 3, 4, \dots$

34. Select the symbol that makes each comparison true.

	<	=	>
$0.25 \square 0.3$	(A)	(B)	(C)
$0.25 \square 0.241$	(D)	(E)	(F)
$0.25 \square 0.250$	(G)	(H)	(I)

35. What is $17.304 + 8.9$?

- Ⓐ 17.393
- Ⓑ 18.194
- Ⓒ 25.204
- Ⓓ 26.204



Grade 5 FAST Mathematics

Test Release: 2025

Grade 5 FAST Mathematics Test Release Support Document

1. Select all the expressions that represent 158.901.

- ☐ (A) 158 ones + 901 tenths
- ☒ (B) 158 ones + 901 thousandths
- ☐ (C) 158 hundreds + 901 hundredths
- ☐ (D) 1 hundred + 58 tens + 9 tenths + 1 thousandth
- ☒ (E) 1 hundred + 58 ones + 90 hundredths + 1 thousandth
- ☒ (F) 1 hundred + 5 tens + 8 ones + 9 tenths + 1 thousandth

Answer Key: B, E, F

Percentage of Students Answering Correctly: 30%

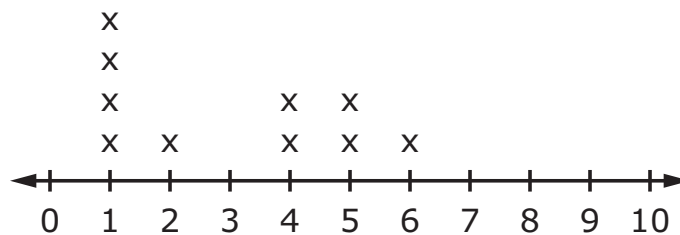
Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.3

Benchmark Description: Compose and decompose multi-digit numbers with decimals to the thousandths in multiple ways using the values of the digits in each place. Demonstrate the compositions or decompositions using objects, drawings and expressions or equations.

Grade 5 FAST Mathematics Test Release Support Document

2. A data set is shown on the line plot.



Select the mean and range of the data set.

	1	3	5
Mean	(A)	(B)	(C)
Range	(D)	(E)	(F)

Answer Key: B, F

Percentage of Students Answering Correctly: 36%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.DP.1.2

Benchmark Description: Interpret numerical data, with whole-number values, represented with tables or line plots by determining the mean, mode, median or range.

3. What is 437×285 ?

124545

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

Answer Key: 124,545, or any equivalent value

Percentage of Students Answering Correctly: 34%

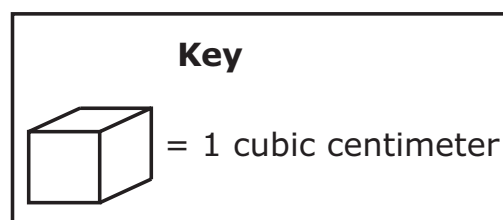
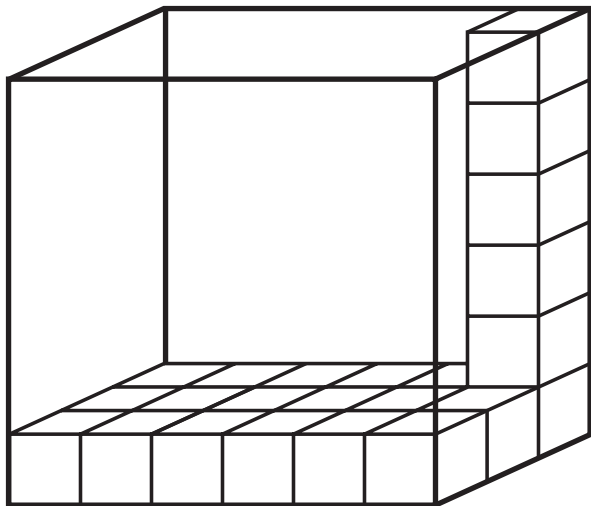
Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.2.1

Benchmark Description: Multiply multi-digit whole numbers including using a standard algorithm with procedural fluency.

4. This question has **two** parts.

A right rectangular prism is shown. The base is covered with cubes with edge lengths of 1 centimeter.



Part A

What is the total volume, in cubic centimeters, of the right rectangular prism?

108

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

Part B

How many more cubes would it take to fill the prism?

85

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: 108, 85, or any equivalent values

Percentage of Students Answering Correctly: 25%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.3.1

Benchmark Description: Explore volume as an attribute of three-dimensional figures by packing them with unit cubes without gaps. Find the volume of a right rectangular prism with whole-number side lengths by counting unit cubes.

5. An input-output table models the expression $3x - 1$.

Input (x)	1	2	4
Output	2	5	11

Select all the columns that could be added to the table.

(A)

Input (x)	7
Output	22

(C)

Input (x)	5
Output	12

(E)

Input (x)	8
Output	23

(B)

Input (x)	3
Output	8

(D)

Input (x)	6
Output	19

Answer Key: B, E

Percentage of Students Answering Correctly: 45%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.3.2

Benchmark Description: Given a rule for a numerical pattern, use a two-column table to record the inputs and outputs.

Grade 5 FAST Mathematics Test Release Support Document

6. Select a word and a phrase to complete the sentence describing attributes of a rhombus.

Since a rhombus is always a

☐ A square
☐ B rectangle
☒ C parallelogram

 ,

it must have

☐ A two long sides and two short sides
☐ B four equal angles
☒ C two pairs of parallel sides

 .

Answer Key: C, C

Percentage of Students Answering Correctly: 60%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.1.1

Benchmark Description: Classify triangles or quadrilaterals into different categories based on shared defining attributes. Explain why a triangle or quadrilateral would or would not belong to a category.

Grade 5 FAST Mathematics Test Release Support Document

7. Natalia has 2 packages of pencils. There are 10 pencils in each package. She gives away some pencils, p , to her friends. She has 8 pencils remaining.

Which equation can be used to determine the number of pencils, p , Natalia gives to her friends?

- Ⓐ $(p - 8) \times 2 = 10$
- Ⓑ $10 \times 2 - p = 8$
- Ⓒ $8 \times 2 - 10 = p$
- Ⓓ $10 \times 2 + 8 = p$

Answer Key: B

Percentage of Students Answering Correctly: 83%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.2.4

Benchmark Description: Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.

8. Katie walks 2 miles. Every $\frac{1}{2}$ mile, she stops to stretch.

Which expression represents how many times Katie stops to stretch on her walk?

Ⓐ $2 \times \frac{1}{2}$

Ⓑ $2 \div \frac{1}{2}$

Ⓒ $2 + \frac{1}{2}$

Ⓓ $2 - \frac{1}{2}$

Answer Key: B

Percentage of Students Answering Correctly: 51%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.AR.1.3

Benchmark Description: Solve real-world problems involving division of a unit fraction by a whole number and a whole number by a unit fraction.

9. What is $11,570 \div 43$?

Ⓐ $245 \frac{35}{43}$

Ⓑ $260 \frac{39}{43}$

Ⓒ $269 \frac{3}{43}$

Ⓓ $273 \frac{38}{43}$

Answer Key: C

Percentage of Students Answering Correctly: 56%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.2.2

Benchmark Description: Divide multi-digit whole numbers, up to five digits by two digits, including using a standard algorithm with procedural fluency. Represent remainders as fractions.

10. The number 0.307 is incorrectly written in expanded form, as shown.

$$3 \times \frac{1}{10} + 7 \times \frac{1}{100}$$

Which action will correct the error?

- Ⓐ replacing $3 \times \frac{1}{10}$ with $30 \times \frac{1}{10}$
- Ⓑ replacing $3 \times \frac{1}{10}$ with $3 \times \frac{1}{1000}$
- Ⓒ replacing $7 \times \frac{1}{100}$ with $70 \times \frac{1}{10}$
- Ⓓ replacing $7 \times \frac{1}{100}$ with $7 \times \frac{1}{1000}$

Answer Key: D

Percentage of Students Answering Correctly: 55%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.2

Benchmark Description: Read and write multi-digit numbers with decimals to the thousandths using standard form, word form and expanded form.

Grade 5 FAST Mathematics Test Release Support Document

11. A partially completed input-output table for the expression $4x$ is shown.

Input (x)	1	3	8	?
Output	4	12	32	48

What is the value of the missing input?

12

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: 12, or any equivalent value

Percentage of Students Answering Correctly: 74%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.3.2

Benchmark Description: Given a rule for a numerical pattern, use a two-column table to record the inputs and outputs.

12. Which comparison is true?

- Ⓐ $367.115 > 376.114$
- Ⓑ $367.115 > 367.105$
- Ⓒ $367.115 > 376.015$
- Ⓓ $367.115 > 367.511$

Answer Key: B

Percentage of Students Answering Correctly: 82%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.4

Benchmark Description: Plot, order and compare multi-digit numbers with decimals up to the thousandths.

13. What is the value of $54.2 \div 0.1$?

A digital calculator interface. At the top, a display box shows the number 542 in red. Below the display is a row of five circular buttons: a left arrow, a right arrow, a circular arrow, a square arrow, and a button with an 'X' in a circle. Below these buttons is a grid of buttons for digits and operations. The grid has four rows and three columns. The first row contains buttons for 1, 2, and 3. The second row contains buttons for 4, 5, and 6. The third row contains buttons for 7, 8, and 9. The fourth row contains buttons for 0, a decimal point (.), and a fraction template button (represented by a horizontal line with a box above and below it). The entire calculator interface is enclosed in a light gray border.

Answer Key: 542, or any equivalent decimal value

Percentage of Students Answering Correctly: 35%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.NSO.2.5

Benchmark Description: Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.

14. An expression is shown.

$$\frac{13}{9} - \frac{3}{4}$$

What is the value of the expression?

Ⓐ $\frac{10}{5}$

Ⓑ $\frac{25}{5}$

Ⓒ $\frac{10}{36}$

Ⓓ $\frac{25}{36}$

Answer Key: D

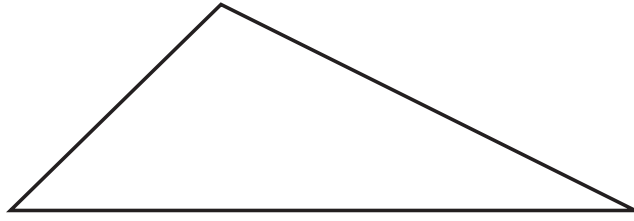
Percentage of Students Answering Correctly: 51%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.FR.2.1

Benchmark Description: Add and subtract fractions with unlike denominators, including mixed numbers and fractions greater than 1, with procedural reliability.

15. A triangle is shown.



Which pair of terms classifies the triangle?

- Ⓐ acute and scalene
- Ⓑ acute and isosceles
- Ⓒ obtuse and scalene
- Ⓓ obtuse and isosceles

Answer Key: C

Percentage of Students Answering Correctly: 32%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.1.1

Benchmark Description: Classify triangles or quadrilaterals into different categories based on shared defining attributes. Explain why a triangle or quadrilateral would or would not belong to a category.

- 16.** Susanna selects a recipe that uses $\frac{5}{3}$ teaspoons of salt. She has $\frac{1}{2}$ teaspoon of salt.

How much more salt does Susanna need for the recipe?

- Ⓐ $\frac{4}{1}$ teaspoons
- Ⓑ $\frac{6}{5}$ teaspoons
- Ⓒ $\frac{7}{6}$ teaspoons
- Ⓓ $\frac{13}{6}$ teaspoons

Answer Key: C

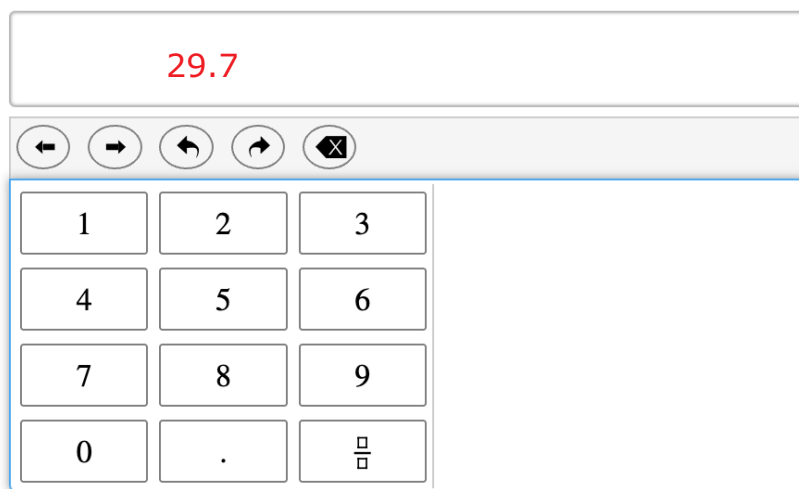
Percentage of Students Answering Correctly: 44%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.AR.1.2

Benchmark Description: Solve real-world problems involving the addition, subtraction or multiplication of fractions, including mixed numbers and fractions greater than 1.

17. What is the value of $81.128 - 51.428$?



A digital calculator interface. The top display shows the number 29.7 in red. Below the display is a toolbar with five circular icons: a left arrow, a right arrow, a circular arrow, a square arrow, and a square with an 'X'. Below the toolbar is a numeric keypad with a 4x3 grid of buttons. The buttons are labeled 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, a decimal point, and a fraction template icon (a square with a horizontal line and two smaller squares above and below it). The keypad is highlighted with a blue border.

Answer Key: 29.7, or any equivalent value

Percentage of Students Answering Correctly: 58%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.NSO.2.3

Benchmark Description: Add and subtract multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.

Grade 5 FAST Mathematics Test Release Support Document

18. A library has 5 large boxes that contain books for a book sale.

- Each box contains 24 hardcover books and 32 paperback books.
- The library sells 77 books on the first day of the book sale.

How many books does the library have left after the first day of the book sale?

203

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: 203, or any equivalent value

Percentage of Students Answering Correctly: 45%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.1.1

Benchmark Description: Solve multi-step real-world problems involving any combination of the four operations with whole numbers, including problems in which remainders must be interpreted within the context.

19. Which equation is true?

Ⓐ $4.2 \div (2 + 5) = 1.50 + 1.44$

Ⓑ $4.2 \div (2 + 5) = 1.50 - 1.44$

Ⓒ $4.2 \div (2 + 5) = 3.85 + 3.25$

Ⓓ $4.2 \div (2 + 5) = 3.85 - 3.25$

Answer Key: D

Percentage of Students Answering Correctly: 31%

Reporting Category: Algebraic Reasoning

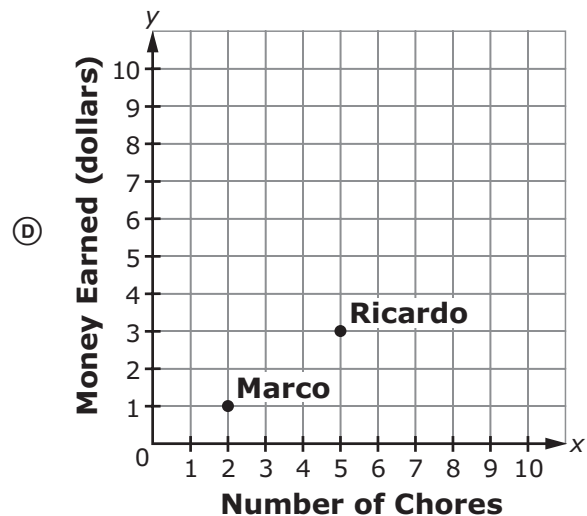
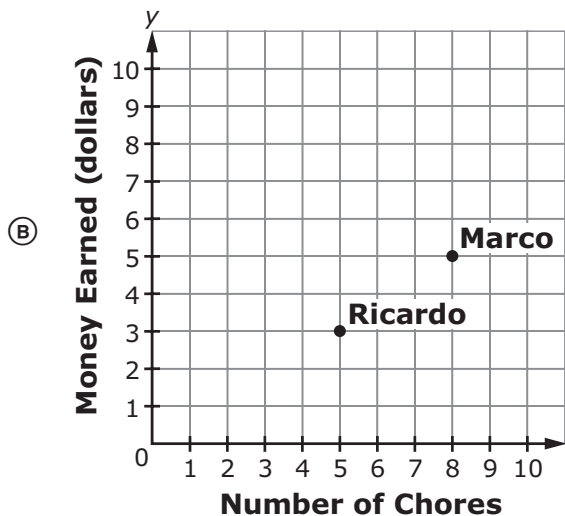
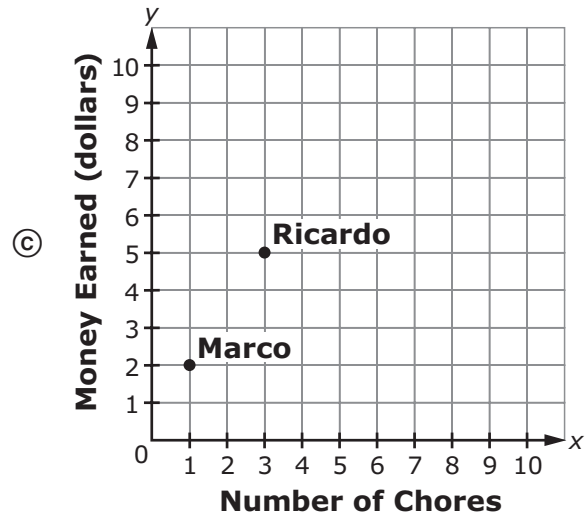
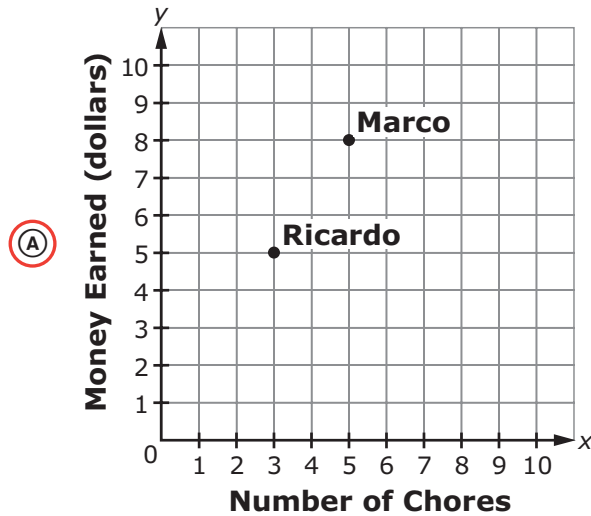
Benchmark: MA.5.AR.2.3

Benchmark Description: Determine and explain whether an equation involving any of the four operations is true or false.

Grade 5 FAST Mathematics Test Release Support Document

- 20.** Ricardo completes 3 chores and earns 5 dollars. Marco completes 2 more chores and earns 3 more dollars than Ricardo. They plot points on a graph to show how many chores and how much money they each earn.

Which graph shows the points for Ricardo and Marco?



Answer Key: A

Percentage of Students Answering Correctly: 78%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.4.2

Benchmark Description: Represent mathematical and real-world problems by plotting points in the first quadrant of the coordinate plane and interpret coordinate values of points in the context of the situation.

21. An expression is shown.

$$3 + 56 \div 8$$

What is the value of the expression?

10

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: 10, or any equivalent value

Percentage of Students Answering Correctly: 60%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.2.2

Benchmark Description: Evaluate multi-step numerical expressions using order of operations.

22. A number is shown in expanded form.

$$(9 \times 100) + (4 \times 10) + (8 \times 1) + \left(2 \times \frac{1}{10}\right) + \left(5 \times \frac{1}{100}\right)$$

What is the number in standard decimal form?

948.25

← → ↶ ↷ ✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\square}{\square}$

Answer Key: 948.25, or any equivalent decimal

Percentage of Students Answering Correctly: 63%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.2

Benchmark Description: Read and write multi-digit numbers with decimals to the thousandths using standard form, word form and expanded form.

Grade 5 FAST Mathematics Test Release Support Document

- 23.** A library has 649 fiction books and 506 nonfiction books. All the books are on 11 shelves. Each shelf has the same number of books.

How many books are on each shelf?

- Ⓐ 15
- Ⓑ 46
- Ⓒ 59
- Ⓓ 105

Answer Key: D

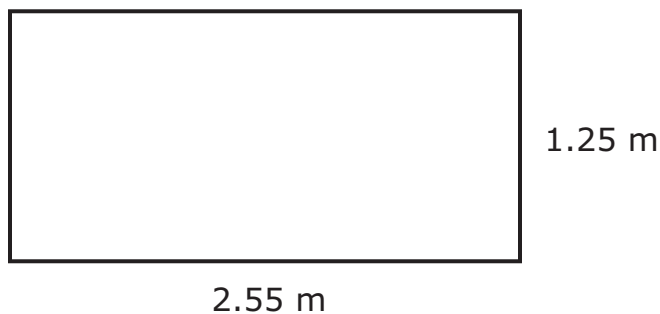
Percentage of Students Answering Correctly: 66%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.1.1

Benchmark Description: Solve multi-step real-world problems involving any combination of the four operations with whole numbers, including problems in which remainders must be interpreted within the context.

24. The rectangle shown has side lengths in meters (m).



What is the perimeter, in meters, of the rectangle?

7.6

←
→
↶
↷
✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: 7.6, or any equivalent value

Percentage of Students Answering Correctly: 60%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.2.1

Benchmark Description: Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

Grade 5 FAST Mathematics Test Release Support Document

- 25.** Cheryl has $8\frac{2}{5}$ pounds of apples. She uses $4\frac{1}{2}$ pounds to make applesauce and $3\frac{1}{4}$ pounds to make apple pie.

How many pounds of apples does Cheryl have left over?

$$\frac{13}{20}$$

←

→

↶

↷

✖

1	2	3
4	5	6
7	8	9
0	.	$\frac{\Box}{\Box}$

Answer Key: $\frac{13}{20}$, or any equivalent value

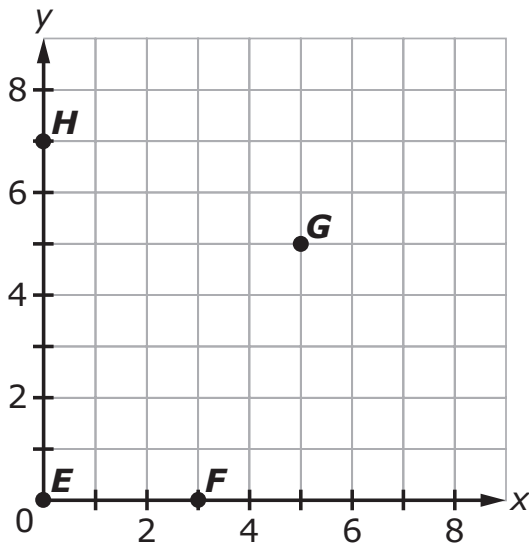
Percentage of Students Answering Correctly: 25%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.AR.1.2

Benchmark Description: Solve real-world problems involving the addition, subtraction or multiplication of fractions, including mixed numbers and fractions greater than 1.

26. Four points are shown on the coordinate plane.



Match each point to its location.

	x-Axis Only	y-Axis Only	Origin	Neither Axis nor Origin
point E	(A)	(B)	(C)	(D)
point F	(E)	(F)	(G)	(H)
point G	(I)	(J)	(K)	(L)
point H	(M)	(N)	(O)	(P)

Answer Key: C, E, L, N

Percentage of Students Answering Correctly: 56%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.4.1

Benchmark Description: Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.

27. Which equation is true?

- Ⓐ $5 + 3 \times 4 = 4 \times 3 + 5$
- Ⓑ $5 + 3 \times 4 = 5 \times 3 + 4$
- Ⓒ $5 + 3 \times 4 = (5 + 3) \times 4$
- Ⓓ $5 + 3 \times 4 = 5 \times 4 + 3 \times 4$

Answer Key: A

Percentage of Students Answering Correctly: 59%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.2.3

Benchmark Description: Determine and explain whether an equation involving any of the four operations is true or false.

28. Select all the numbers that have a value of 38.5 when rounded to the nearest tenth.

☒ (A) 38.518

☒ (B) 38.464

☐ (C) 38.553

☐ (D) 38.417

☒ (E) 38.507

Answer Key: A, B, E

Percentage of Students Answering Correctly: 53%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.5

Benchmark Description: Round multi-digit numbers with decimals to the thousandths to the nearest hundredth, tenth or whole number.

29. What is $2,372 \times 431$?

- Ⓐ 889,232
- Ⓑ 900,332
- Ⓒ 1,022,332
- Ⓓ 10,223,320

Answer Key: C

Percentage of Students Answering Correctly: 66%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.2.1

Benchmark Description: Multiply multi-digit whole numbers including using a standard algorithm with procedural fluency.

30. What is the value of $\frac{30}{12} + \frac{4}{8}$?

Ⓐ $\frac{34}{24}$

Ⓑ $\frac{34}{20}$

Ⓒ $\frac{72}{24}$

Ⓓ $\frac{38}{12}$

Answer Key: C

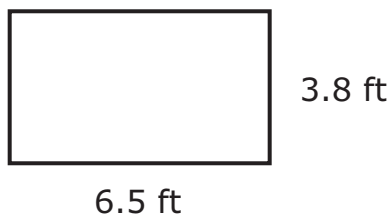
Percentage of Students Answering Correctly: 51%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.FR.2.1

Benchmark Description: Add and subtract fractions with unlike denominators, including mixed numbers and fractions greater than 1, with procedural reliability.

31. A rectangle is shown, with side lengths in feet (ft).



Complete the sentences to show the perimeter and area of the rectangle.

The perimeter is

20.6

feet.

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
0	.	$\frac{\Box}{\Box}$		

The area is

24.7

square feet.

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
0	.	$\frac{\Box}{\Box}$		

Grade 5 FAST Mathematics Test Release Support Document

Answer Key: 20.6, 24.7, or any equivalent values

Percentage of Students Answering Correctly: 42%

Reporting Category: Geometric Reasoning, Measurement, and Data Analysis and Probability

Benchmark: MA.5.GR.2.1

Benchmark Description: Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

32. A statement is shown.

“The sum of 37 and the product of 5 and p is 97.”

Which equation represents the statement?

- ☒ Ⓐ $37 + (5 \times p) = 97$
- ☐ Ⓑ $37 \times (5 + p) = 97$
- ☐ Ⓒ $(37 + 5) \times p = 97$
- ☐ Ⓓ $(37 \times 5) + p = 97$

Answer Key: A

Percentage of Students Answering Correctly: 72%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.2.4

Benchmark Description: Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.

33. Alex writes the pattern shown.

5, 8, 11, 14, . . .

He wants to create a rule that describes the pattern.

Select terms to complete the statement describing a rule for Alex's pattern.

A rule that describes the pattern is [☐ A $2x$ ☒ B $3x$ ☐ C $5x$] +
[☐ A 0 ☒ B 2 ☐ C 3 ☐ D 5] , where $x = 1, 2, 3, 4, . . .$

Answer Key: B, B

Percentage of Students Answering Correctly: 24%

Reporting Category: Algebraic Reasoning

Benchmark: MA.5.AR.3.1

Benchmark Description: Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.

Grade 5 FAST Mathematics Test Release Support Document

34. Select the symbol that makes each comparison true.

	<	=	>
$0.25 \square 0.3$	☒ (A)	☐ (B)	☐ (C)
$0.25 \square 0.241$	☐ (D)	☐ (E)	☒ (F)
$0.25 \square 0.250$	☐ (G)	☒ (H)	☐ (I)

Answer Key: A, F, H

Percentage of Students Answering Correctly: 65%

Reporting Category: Number Sense and Operations with Whole Numbers

Benchmark: MA.5.NSO.1.4

Benchmark Description: Plot, order and compare multi-digit numbers with decimals up to the thousandths.

35. What is $17.304 + 8.9$?

- Ⓐ 17.393
- Ⓑ 18.194
- Ⓒ 25.204
- Ⓓ 26.204

Answer Key: D

Percentage of Students Answering Correctly: 78%

Reporting Category: Number Sense and Operations with Fractions and Decimals

Benchmark: MA.5.NSO.2.3

Benchmark Description: Add and subtract multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.